

ODE Systems Terminology

*First-Order Systems, Matrix-Vector Notation, State Variables
& Translating Higher-Order Scalar Equations*

Prepared by Staples Education

First-Order Systems • State Variables • State Vector • Matrix-
Vector Form • Companion Matrix • Higher-Order $\rightarrow$ System •
Coupled & Variable-Coefficient Systems • The Reverse Direction

Contents

1	Part I Condensed Cheat Sheet	2
1.1	First-Order Systems: The Basic Object	2
1.2	State Variables and the State Vector	2
1.3	Matrix-Vector Notation	2
1.4	Translating a Higher-Order Scalar ODE into a System	3
1.5	Practical Translation Cases	4
1.6	The Reverse Direction & Why We Convert	4
2	Part II Review Guide	6
3	Part III Answer Key & Worked Solutions	7

1 Part I | Condensed Cheat Sheet

1.1 First-Order Systems: The Basic Object

Most of the theory of differential equations is built on a single canonical object — the *first-order system*, in which every equation involves only first derivatives of the unknowns. Higher-order equations, coupled equations, and mechanical models all reduce to this one form, which is exactly why it is worth naming its parts carefully.

General first-order system (n unknowns $x_1, \dots, x_n$):

$$x'_1 = f_1(t, x_1, \dots, x_n), \quad \dots, \quad x'_n = f_n(t, x_1, \dots, x_n).$$

Vocabulary.

- *State variables* — the unknowns $x_1, \dots, x_n$.
- *Dimension / order* — the number n of equations (and of state variables).
- *Linear* — each f_i is linear in $x_1, \dots, x_n$.
- *Autonomous* — no t appears explicitly on the right-hand side.
- *Homogeneous* (linear case) — no stand-alone forcing term.

1.2 State Variables and the State Vector

The state variables are stacked into a single *state vector* $\mathbf{x}(t) = (x_1, \dots, x_n)^\top$. The defining property of a state is sufficiency — knowing $\mathbf{x}(t_0)$ determines the entire future of the system. For a moving mass, “where it is” alone is not a state, but “position together with velocity” is.

1.3 Matrix–Vector Notation

When the system is linear, all the coefficients collect into a matrix and the whole system compresses to one line.

Linear system in matrix form:

$$\mathbf{x}' = P(t) \mathbf{x} + \mathbf{g}(t),$$

where $P(t)$ is the $n \times n$ *coefficient matrix* and $\mathbf{g}(t)$ the *forcing (inhomogeneity) vector*.

- *Homogeneous*: $\mathbf{g}(t) = \mathbf{0} \Rightarrow \mathbf{x}' = P(t)\mathbf{x}$.
- *Constant-coefficient*: $P(t) = A$ constant $\Rightarrow \mathbf{x}' = A\mathbf{x} + \mathbf{g}(t)$.
- *IVP*: pair the system with $\mathbf{x}(t_0) = \mathbf{x}^0$.

Existence/uniqueness. If $P(t)$ and $\mathbf{g}(t)$ are continuous on an interval containing t_0 , the IVP has exactly one solution there.

Worked Example — reading off A and $\mathbf{g}$. Write $x'_1 = 3x_1 - 2x_2 + e^t$, $x'_2 = x_1 + 4x_2$ in matrix form.

Step 1. Collect the coefficients of x_1, x_2 row by row:

$$A = \begin{pmatrix} 3 & -2 \\ 1 & 4 \end{pmatrix}.$$

Step 2. Collect the terms with no state variable into $\mathbf{g}$:

$$\mathbf{g}(t) = \begin{pmatrix} e^t \\ 0 \end{pmatrix}.$$

Step 3. Assemble:

$$\mathbf{x}' = \begin{pmatrix} 3 & -2 \\ 1 & 4 \end{pmatrix} \mathbf{x} + \begin{pmatrix} e^t \\ 0 \end{pmatrix}.$$

This system is linear, constant-coefficient, and non-homogeneous.

1.4 Translating a Higher-Order Scalar ODE into a System

This is the central skill — and the reason first-order systems are universal. Numerical solvers, eigenvalue theory, and phase-plane analysis all require first-order form, so any scalar equation of order n must first be unpacked into n first-order equations.

The recipe. Given $y^{(n)} = F(t, y, y', \dots, y^{(n-1)})$, define the state variables as the successive derivatives:

$$x_1 = y, \quad x_2 = y', \quad x_3 = y'', \quad \dots, \quad x_n = y^{(n-1)}.$$

Then the system is

$$x'_1 = x_2, \quad x'_2 = x_3, \quad \dots, \quad x'_{n-1} = x_n, \quad x'_n = F(t, x_1, \dots, x_n).$$

Linear case. For $y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_1y' + a_0y = g(t)$, the matrix is the *companion matrix*:

$$A = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{pmatrix}, \quad \mathbf{g} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ g(t) \end{pmatrix}.$$

Always normalize first — divide through so the leading coefficient is 1 before reading off the a_i .

Worked Example — second order to a system. Convert $y'' + 5y' + 6y = 0$.

Step 1. Define states:

$$x_1 = y, \quad x_2 = y'.$$

Step 2. The first equation is automatic; the second solves for the top derivative:

$$x'_1 = x_2, \quad x'_2 = y'' = -6y - 5y' = -6x_1 - 5x_2.$$

Step 3. Matrix form:

$$\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix} \mathbf{x}.$$

Worked Example — third order, forced. Convert $y''' - 2y'' + y' - 3y = \cos t$.

Step 1. Define three states:

$$x_1 = y, \quad x_2 = y', \quad x_3 = y''.$$

Step 2. Solve for the highest derivative and substitute:

$$x_3' = y''' = 3y - y' + 2y'' + \cos t = 3x_1 - x_2 + 2x_3 + \cos t.$$

Step 3. Assemble the chain plus the companion row:

$$\mathbf{x}' = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 3 & -1 & 2 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 0 \\ 0 \\ \cos t \end{pmatrix}.$$

1.5 Practical Translation Cases

Three situations come up constantly — a leading mass coefficient, variable coefficients, and several coupled equations. The recipe is the same; only the bookkeeping changes.

Mechanical systems. A spring–mass–damper $mu'' + \gamma u' + ku = F(t)$ becomes a first-order system in the physical state (position, velocity).

Worked Example — spring–mass–damper to state space. Recast $mu'' + \gamma u' + ku = F(t)$.

Step 1. Normalize (divide by m) and choose physical states:

$$x_1 = u \text{ (position)}, \quad x_2 = u' \text{ (velocity)}.$$

Step 2. The acceleration equation:

$$x_2' = u'' = -\frac{k}{m}u - \frac{\gamma}{m}u' + \frac{F(t)}{m} = -\frac{k}{m}x_1 - \frac{\gamma}{m}x_2 + \frac{F(t)}{m}.$$

Step 3. Matrix form:

$$\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -k/m & -\gamma/m \end{pmatrix} \mathbf{x} + \begin{pmatrix} 0 \\ F(t)/m \end{pmatrix}.$$

Variable coefficients produce a t -dependent matrix $P(t)$ — still a perfectly valid first-order system, just not constant-coefficient. **Coupled higher-order equations** (e.g. two masses) are handled by giving *each* dependent variable its own position-and-velocity pair, so two second-order equations become a four-dimensional first-order system.

1.6 The Reverse Direction & Why We Convert

The translation also runs backward — a linear system can be collapsed into a single higher-order scalar ODE by *elimination*: differentiate one equation, then use the others to remove the extra unknowns.

Note (Consistency Check). When a constant-coefficient system $\mathbf{x}' = A\mathbf{x}$ is collapsed to a scalar equation, that scalar equation's characteristic polynomial equals $\det(A - \lambda I)$ — the eigenvalues of A are precisely the characteristic roots. This is the bridge between the matrix picture and the scalar picture.

We convert because first-order systems are the common language of the subject: numerical methods (Euler, Runge–Kutta) integrate $\mathbf{x}' = \mathbf{f}(t, \mathbf{x})$ directly, and the qualitative tools — eigenvalues, stability, phase portraits — are all stated for systems.

2 Part II | Review Guide

Ten problems, ordered from terminology to multi-equation translation. Full solutions follow in Part III.

1. Consider the system $x_1' = 2x_1 - x_2 + t$, $x_2' = 4x_1 + 3x_2$. (a) Name the state variables and the dimension. (b) Is it linear? (c) Homogeneous? (d) Autonomous? (e) Write it in matrix form.
2. Convert the scalar equation $y'' + 4y = 0$ into a first-order system. State the state variables and the two first-order equations.
3. Write the system from Problem 2 in matrix form $\mathbf{x}' = A\mathbf{x}$, giving the matrix A .
4. Convert $y'' - 3y' + 2y = e^t$ into a first-order system in matrix form $\mathbf{x}' = A\mathbf{x} + \mathbf{g}(t)$. Identify A and $\mathbf{g}(t)$.
5. Convert the third-order equation $y''' + 2y'' - y' + 5y = 0$ into a first-order system, and write its companion matrix.
6. A damped spring-mass system satisfies $2u'' + 3u' + 8u = 4\cos(3t)$. Using $x_1 = u$, $x_2 = u'$, write the equivalent first-order system in matrix form.
7. Convert the variable-coefficient equation $t^2y'' - ty' + y = 0$ (for $t > 0$) into a first-order system $\mathbf{x}' = P(t)\mathbf{x}$. Give $P(t)$ and state why the system is not constant-coefficient.
8. Two coupled oscillators satisfy $x'' = -2x + y$ and $y'' = x - 2y$. Introduce state variables and write the equivalent first-order system in matrix form. What is its dimension?
9. Given the first-order system $x_1' = x_1 + x_2$, $x_2' = 4x_1 + x_2$, eliminate x_2 to recover a single second-order scalar ODE for x_1 .
10. The undamped pendulum obeys $\theta'' + \frac{g}{L}\sin\theta = 0$. (a) Write it as a first-order system using $x_1 = \theta$, $x_2 = \theta'$. (b) Explain why it cannot be written as $\mathbf{x}' = A\mathbf{x}$ with constant A . (c) Give the small-angle linearization in matrix form.

3 Part III | Answer Key & Worked Solutions

Solution 1

- (a) The state variables are x_1 and x_2 ; the dimension (order) is 2.
- (b) Linear — every right-hand side is a linear combination of x_1, x_2 (plus a function of t).
- (c) Non-homogeneous — the term $+t$ stands alone, not multiplying any state variable.
- (d) Non-autonomous — the variable t appears explicitly on the right.
- (e) Matrix form:

$$\mathbf{x}' = \begin{pmatrix} 2 & -1 \\ 4 & 3 \end{pmatrix} \mathbf{x} + \begin{pmatrix} t \\ 0 \end{pmatrix}.$$

Solution 2

Define the state variables as y and its first derivative:

$$x_1 = y, \quad x_2 = y'.$$

The chain rule gives the first equation automatically; solving $y'' = -4y$ gives the second:

$$x_1' = x_2, \quad x_2' = y'' = -4y = -4x_1.$$

Solution 3

Reading the coefficients off Solution 2:

$$A = \begin{pmatrix} 0 & 1 \\ -4 & 0 \end{pmatrix}, \quad \mathbf{x}' = A\mathbf{x}.$$

Solution 4

States $x_1 = y$, $x_2 = y'$. Solve for the top derivative: $y'' = 3y' - 2y + e^t$. Then

$$x_1' = x_2, \quad x_2' = -2x_1 + 3x_2 + e^t.$$

$$A = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}, \quad \mathbf{g}(t) = \begin{pmatrix} 0 \\ e^t \end{pmatrix}.$$

Solution 5

States $x_1 = y$, $x_2 = y'$, $x_3 = y''$. Solving for the top derivative: $y''' = -5y + y' - 2y''$. The system:

$$x_1' = x_2, \quad x_2' = x_3, \quad x_3' = -5x_1 + x_2 - 2x_3.$$

Companion matrix:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5 & 1 & -2 \end{pmatrix}.$$

(The bottom row is $(-a_0, -a_1, -a_2) = (-5, 1, -2)$, matching $a_0 = 5$, $a_1 = -1$, $a_2 = 2$.)

Solution 6

Normalize by dividing by the mass $m = 2$:

$$u'' = -4u - \frac{3}{2}u' + 2\cos(3t).$$

With $x_1 = u$, $x_2 = u'$:

$$\begin{aligned} x_1' &= x_2, & x_2' &= -4x_1 - \frac{3}{2}x_2 + 2\cos(3t). \\ \mathbf{x}' &= \begin{pmatrix} 0 & 1 \\ -4 & -\frac{3}{2} \end{pmatrix} \mathbf{x} + \begin{pmatrix} 0 \\ 2\cos(3t) \end{pmatrix}. \end{aligned}$$

The forgotten division by m is the most common error here — the entries must come from the *normalized* equation.

Solution 7

Put the equation in standard form by dividing by t^2 (valid since $t > 0$):

$$y'' - \frac{1}{t}y' + \frac{1}{t^2}y = 0 \implies y'' = \frac{1}{t}y' - \frac{1}{t^2}y.$$

With $x_1 = y$, $x_2 = y'$:

$$\begin{aligned} x_1' &= x_2, & x_2' &= -\frac{1}{t^2}x_1 + \frac{1}{t}x_2. \\ \mathbf{x}' &= P(t)\mathbf{x}, & P(t) &= \begin{pmatrix} 0 & 1 \\ -1/t^2 & 1/t \end{pmatrix}. \end{aligned}$$

The entries of P depend on t — so the system is linear but *variable-coefficient*, not constant-coefficient.

Solution 8

Each dependent variable needs a position–velocity pair. Set

$$z_1 = x, \quad z_2 = x', \quad z_3 = y, \quad z_4 = y'.$$

Then, using $x'' = -2x + y$ and $y'' = x - 2y$:

$$z_1' = z_2, \quad z_2' = -2z_1 + z_3, \quad z_3' = z_4, \quad z_4' = z_1 - 2z_3.$$

$$\mathbf{z}' = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & -2 & 0 \end{pmatrix} \mathbf{z}.$$

Two second-order equations yield a system of dimension 4.

Solution 9

From the first equation, solve for the extra unknown:

$$x_2 = x_1' - x_1.$$

Differentiate the first equation and substitute the second:

$$x_1'' = x_1' + x_2' = x_1' + (4x_1 + x_2) = x_1' + 4x_1 + (x_1' - x_1) = 2x_1' + 3x_1.$$

Rearranging gives the single second-order ODE:

$$x_1'' - 2x_1' - 3x_1 = 0.$$

Check: the matrix $\begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}$ has $\text{tr} = 2$, $\det = -3$, so its characteristic equation is $\lambda^2 - 2\lambda - 3 = 0$ — identical to the scalar equation, as expected.

Solution 10

(a) With $x_1 = \theta$ (angle) and $x_2 = \theta'$ (angular velocity):

$$x_1' = x_2, \quad x_2' = \theta'' = -\frac{g}{L} \sin x_1.$$

(b) The term $\sin x_1$ is a nonlinear function of the state, so the right-hand side cannot be written as a constant matrix times $\mathbf{x}$ — the system is *nonlinear*.

(c) For small angles, $\sin x_1 \approx x_1$, giving the linearization

$$\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -g/L & 0 \end{pmatrix} \mathbf{x}.$$

The coefficient matrix has pure-imaginary eigenvalues $\lambda = \pm i\sqrt{g/L}$, the phase-plane signature of a center — the familiar undamped oscillation.